

Bavly Fayed

Iowa | <https://bfayed.github.io/Bavly-Portfolio> | bavlyfayed@gmail.com | LinkedIn: Bavlyfayed2003

EDUCATION

Iowa State University

Bachelor of Science: Computer Science, GPA: 3.2

Ames, IA

Graduation Date: Dec 2026

WORK EXPERIENCE

AspenTech / Emerson

Software Engineering Intern

Medina, MN

May 2026 - Aug 2026

- Developed full-stack capabilities for a large-scale enterprise software platform using Java, Angular, TypeScript, and RESTful services.
- Designed and implemented metadata ingestion workflows that collect, process, and expose system, profile, configuration, and runtime data across multiple platform components.
- Enhanced product functionality and maintainability by analyzing complex codebases, resolving integration issues, and delivering scalable features across interconnected services.

Altec

Co-op / Full-time Full-stack

Osceola, IA

May 2025 - Dec 2025

- Contributed to the development and maintenance of internal software tools using Angular, TypeScript, and C#/.NET, improving operational efficiency for manufacturing processes.
- Developed and optimized RESTful APIs for communication between frontend components and backend services.
- Utilized SQL Server to query and update production-related data, ensuring data consistency across systems.
- Applied NgRx Component Store for efficient state management across Angular components, improving app scalability and maintainability.
- Designed and created tooling inventory system to manage usage and tooling health.
- Implemented SignalR for live connection notifications and subscribing to other users activity to all users with access.

Vontas

Intern Back-end

Cedar Rapids, IA

Jun 2024 - Aug 2024

- Contributed to the development of a website using Spring Boot, assisting in streamlining backend processes.
- Supported the integration of RabbitMQ for asynchronous communication between services, aiding in the optimization of high-volume data transactions and ensuring message reliability in distributed systems.
- Assisted in implementing an MVC (Model-View-Controller) architecture to ensure a clean separation of concerns, enhancing code maintainability.
- Collaborated with senior developers to enhance application scalability and fault tolerance by utilizing message queuing and microservices architecture.

PROJECTS

Group Project

Android Application

Ames, IA

- Designed and developed a collaborative Android application that enables users to seamlessly manage a shared calendar for scheduling events and appointments.
- Implemented intuitive user interfaces with responsive designs, ensuring a smooth and user-friendly experience across different devices.

Real-Time Wagering Game

- Built a real-time wagering game with 4 live lobbies, spectator mode, and live player chat.
- Designed Spring Boot APIs and SQL-backed services to manage lobbies, track wagers, and handle payouts.
- Used WebSockets to deliver fast live game updates and chat to all connected players.

Portfolio Generator Web-Based Application

- The application automates the creation of personalized portfolio websites by generating professional portfolios based on user inputs.
- The application allows users to customize sections like experience, projects, and contact information dynamically.

SKILLS

Java, Python, Mkdocs, HTML, CSS, JavaScript, Node.js, React, SQL Server, Express.js, Spring Boot, Angular, TypeScript, C#, .NET, Git, RabbitMQ, JSP, MVC, NgRx, Kendo UI, MongoDB, UI/UX.